

ex 13

English Recent Login Sign up

```

1 clear;
2 clc;
3 close all;
4
5 %% PARAMETERS
6
7 frequencies = [28e9 60e9]; % Frequencies in Hz
8
9 c = 3e8; % Speed of light (m/s)
10
11 distances = 1:1:100; % Distance in meters
12
13 epsilon_r_wall = 5; % Relative permittivity of wall
14
15 theta_inc = 45; % Incident angle in degrees
16
17 % Atmospheric attenuation (dB/m)
18 atm_loss = [0.06 0.2]; % 28 GHz and 60 GHz
19
20 %% INITIALIZE ARRAYS
21
22 ReflectionLoss = zeros(length(frequencies), ...
23     length(distances));
24
25 PenetrationLoss = zeros(length(frequencies), ...
26     length(distances));
27
28 AtmosphericLoss = zeros(length(frequencies), ...
29     length(distances));
30
31 %% =====
32 % COMPUTE LOSS COMPONENTS
33 % =====
34
35 for f_idx = 1:length(frequencies)
36     f = frequencies(f_idx);
37

```

Run

Save

Output

Input

```

=====
mmWAVE LOSS COMPONENT ANALYSIS
=====

Frequency: 28 GHz
Reflection Loss = 6.02 dB
Penetration Loss = 5.56 dB
Atmospheric Loss at 100 m = 6.00 dB

```

```

Frequency: 60 GHz
Reflection Loss = 6.02 dB
Penetration Loss = 6.20 dB
Atmospheric Loss at 100 m = 20.00 dB

```

Simulation completed successfully.

[Execution complete with exit code 0]

